N E X U S Z O N E

Portfolio Companion

Product Catalog · Invention Catalog

EDITION 2026

"Where technology meets craft."
A complete inventory of shipped products, published research, and applied-security field work.

Founded by game_nexus_zone

Confidential portfolio companion · Distribute with attribution

TABLE OF CONTENTS

Inside this book

00	About This Companion	Introduction
00.5	Verification & Evidence Wall — click anything to verify	Anchor Index
01	Nexus Zone — the parent brand	Ch. 1
02	LucidCode — code that confesses	Ch. 2
03	Laundry — the legacy-code laundry room	Ch. 3
04	NEXUS-AI — 134-agent fabric + cyber grounding gate	Ch. 4
05	midcine — Arabic radiology AI	Ch. 5
05.5	MEDNEXA — healthcare integration platform (HIS × LIS × PACS)	Ch. 5.5
06	thawani.cc — WhatsApp commerce SaaS	Ch. 6
07	ARIA v9.0 — sovereign AI companion	Ch. 7
08	NEXUS Vigil™ — abandoned-vehicle intelligence	Ch. 8
09	The Security Stack — SecOps · Firm · SENTINEL · VulnAdvisor	Ch. 9
10	Gaming Universe — 6 platforms	Ch. 10
11	Specialized Products — Zwaya Fit · GZ CyberRange · V24	Ch. 11
12	Published Research	Ch. 12
13	Competitive Results — Kaggle	Ch. 13
14	Responsible-Disclosure Track Record	Ch. 14
15	Roadmap Ahead	Ch. 15

INTRODUCTION

00 — About This Companion

This companion is **not a résumé**. A résumé is a one-page document you read in thirty seconds. This companion is what you read *after* the résumé — when you have decided the products deserve closer inspection.

Every product here follows one design rule: **every claim must be verifiable**. Every number is measurable in a running system. Every architectural choice is defended by a public commit, a working demo, or a peer-review-ready research paper.

> *Products are not judged by the demos they run. They are judged by the demos they refuse to run.*

Inside this catalog you will find **twelve shipping products**, **three research papers**, **six Kaggle competitions**, **thirteen responsible-disclosure engagements** (including a Google OSS VRP S0/P0 acceptance and a live-verified CVSS 9.1 Cal.com submission), and **two hire-signal open-source contributions** at PostHog and Supabase. Each product page states what the product does, what makes it different from its category, the technical signature that gives it edge, and the verified numbers behind its claims.

12+	3	13	2
Shipped products	Research papers	Disclosure engagements	OSS hire-signal PRs

VERIFICATION & EVIDENCE WALL

00.5 — Click Anything, Verify Everything

Every claim in this document is externally verifiable. Below is a single index of the public URLs that back the numbers. If a reviewer wants to independently confirm a specific claim, the corresponding link resolves to the artifact — a merged PR, a published paper, a live demo, a Kaggle placement, an accepted bounty report, or a public product landing page.

Open-Source Contributions · 10 organisations, fully public

Hire-signal PRs — click to verify code, tests, and maintainer review

Merged

- **PostHog PR #5083** — opt-in WebView bot detection (merged by @marandaneto, 11-day round trip) · pull/5083 · MERGED
- **Supabase storage PR #1500** — TUS `X-Supabase-Object-Id` response header (merged by @ferhatelmas, closes 2-year #647) · pull/1500 · MERGED

Open PRs under review

- **Drizzle ORM PR #6497** — `useLiveQuery` on `expo-sqlite` subscribes to joined tables; new `SQLiteRelationalQuery.getReferencedTables()` with cycle guard · pull/6497 · in review
- **Sentry-JS PR #25348** — Nuxt pages-router client-plugin unit tests, 5 code paths × 13 cases (pre-blessed by MEMBER @JPeer264) · pull/25348 · in review
- **Sentry Relay PR #6508** — derived trace context prefers the DSC `trace_id`; 84/84 local tests · pull/6508 · in review
- **SvelteKit PR #17432** — `adapter-cloudflare` swaps the Cloudflare platform proxy in place when `wrangler.toml` changes (Vercel bot: "the bug is fixed") · pull/17432 · in review
- **Inngest PR #5004** — single shared TracerProvider via `BatchSpanProcessor`, stops a per-run `OtelTracerProvider` leak · pull/5004 · in review
- **Supabase storage PR #1488** — JWK auto-selection (closes 607-day #629); 25 tests incl. jose round-trip proof · pull/1488 · in review
- **Supabase storage PR #1499** — magic-byte MIME validator closing #639; 61 tests + 2 security-review iterations with `depthfirst-app` (ZIP/OLE2 bypass closures) · pull/1499 · in review
- **Supabase auth PR #2863** — preserve OAuth 2.1 client sessions during MFA step-up · pull/2863 · in review
- **PostHog PR #5244** — rrweb `MutationBuffer.processBufferedMutations` $O(n^2) \rightarrow O(n)$; 500-row shadow-DOM regression · pull/5244 · in review

Maintainer-blessed design proposals

- **Prisma ORM #20678** — ALS-based transparent transaction propagation for `$extends` on the v7 maintenance line; 8-case failure matrix · issue/20678

- **Node.js core #55422** — LTS-only fix for `Buffer.copy` silent corruption past 4 GiB (v22.x + v24.x), zero V8-ABI change · issue/55422
- **Nitro #4471** — conditions-aware `externals.ts` resolve so `@sentry/tanstackstart-react` doesn't crash the Rollup graph; co-designed with @hiranuma's bisect · issue/4471

Published tool

- **apds-hunt v0.1** — reference implementation of the APDS methodology (XOR Rule theorem + 6-step algorithm + CVE-2024-40658 shape self-test), MIT · github.com/61465/apds-hunt · PUBLIC

Accepted security advisories · public GHSA

First-of-their-kind GHSAs on their respective repos

- **Trigger.dev · GHSA-65rr-73jq-6qhf** — accepted by @ChrisArderne 2026-10-07 (CVSS 9.6; 90-day disclosure clock)
- **Documenso · GHSA-2rp7-mrh6-v9m2** — in triage (CVSS 7.6; first-ever GHSA on the repo; rejected-recipient field forgery across 3 tRPC routes)

Published Research & Methodology · public

Peer-review-ready papers + methodology artefacts

- **APDS — Asymmetric Patch Detection for Symmetric APIs** (v1.0, Oct 2026) · formal XOR Rule theorem + 6-step algorithm + retrospective validation on 5 landmark CVEs (BlueBorne, Stagefright, LogJam, V8 CVE-2016-1646, HDR10+ 2024) · reference tool at github.com/61465/apds-hunt
- **Ariadne — HITL Defence for AI Agents** · SSRN 6870178
- **Oblivion Gate — Zero-Trace Fragmentation** · SSRN 6602478

Kaggle · fully public leaderboards

Live-scored placements across 4 tracks

- **Profile** · kaggle.com/abdalrahmanmohamed3
- **Paper Track — Symbolic Ceiling on ARC-AGI-2** · writeup
- **Pokémon TCG α-Rank Study** · writeup

Live product landings · fully public

Products you can open in a browser

- **Nexus Zone parent site** · 61465.github.io/game-zone-hub
- **LucidCode landing** · 61465.github.io/lucid/landing

- **Laundry interactive demo** · 61465.github.io/lucid/landing/laundry.html
- **Thawani commerce SaaS** · thawani.cc (paid tenants live) · **backup landing:** 61465.github.io/thawanidemo · **source:** github.com/61465/Thawani
- **MEDNEXA healthcare integration platform** · **source:** github.com/61465/MEDNEXA (Egyptian HIS × LIS × PACS wiring; HL7v2 + FHIR R4 + DICOM; PDPC Law 151/2020 + HIPAA-aligned)
- **midcine Arabic radiology AI** · **source:** github.com/61465/midcine
- **Golden Arena 518** · 61465.github.io/game-zone-golden
- **Game Zone Archive** · 61465.github.io/game-zone-archive
- **Waraq Reader** · 61465.github.io/read
- **GZ Chat Pro** · 61465.github.io/GAME-ZONE-CHAT
- **Golden Arena Configurator** · 61465.github.io/game-zone-CONFIGURATOR

Responsible-disclosure anchors

Submission IDs — available for HR verification on request

- **Google Android VRP — APDS methodology + CVE-2024-40658 symmetric sibling (CVSS 10.0 Critical)** · Issue `571135577` · In Panel Review · byte-identical un-patched sibling across 6 AOSP branches (android14-qpr3 → android17-release, 25 months of ASB silence); live binary verification from Android 16 QPR1 emulator image
- **Google OSS VRP — Gemini CLI chained RCE** · Issue `563082721` · S0/P0 Accepted
- **Google Cloud VRP — Vertex AI pickle cross-tenant RCE** · Issue `569156158` · AI-triage done, human review
- **Google OSS VRP — ADK MCP token forwarding** · Issue `569269080` · P1 (upgraded in 5 min)
- **Google Cloud VRP — Agent Builder RLHF / BOLA** · Issue `569402823` · AI-triage done, human review
- **Google Cloud VRP — Agent Builder custom-function RCE** · Issue `569575994` · In triage
- **Syfe · Web Cache Deception (CVSS 8.6)** · HackerOne `#3914035` · Accepted
- **Gearset · Excessive GitHub-OAuth scope** · Bugcrowd `0e7d2c90` · Accepted
- **Cal.com · Zero-click PII leak (CVSS 9.1, live-verified on `app.cal.qa`)** · Bugcrowd `30c5dfce`
- **OpenAI Safety · Cross-tenant PII (P1)** · Bugcrowd `cf0acebe`
- **Trend Micro ZDI · Windows sig-verify gap (CVSS 9.0)** · `NexusZone0001`
- **xAI Grok · Cross-model prompt disclosure** · HackerOne `#4024682` · vendor review
- **Shopify · 7-vector chain (CVSS 8.7)** · HackerOne `#3919475` · in triage
- **AXIS OS · ACAP D-Bus bypass of CVE-2025-5452 (P2)** · Bugcrowd `8a7425f0`
- **eToro · 1-click ATO on Delta Desktop (P1)** · Bugcrowd `f83be6be`
- **Okta AtSpoke · Application-wide CSRF (P1)** · Bugcrowd `6d26be3a`
- **Magic Labs · reCAPTCHA bypass + user enum (P3)** · Bugcrowd `696307c8`

Bugcrowd / HackerOne submission pages require program-side authorization to view. Full triager conversations, PoC videos, patch drafts, and evidence bundles are available under NDA on request.

Kaggle certificates / profile links

Additional identity anchors

- **GitHub** · github.com/61465
- **SSRN author page** · SSRN author 11211189
- **Portfolio Companion source (this document)** · github.com/61465/Nexus_Zone_Portfolio_Companion
- **Contact** · `abdarahman10555@gmail.com` · `+20 120 853 1520`

CHAPTER 1 · THE PARENT BRAND

01 — Nexus Zone

Nexus Zone is the umbrella brand under which every product in this catalog ships. The name unifies two heritages:

- **Game Zone** — the original mastery territory: interactive experiences, gamer platforms, community tools.
- **Nexus** — the connection point that links gaming to AI, cybersecurity, and software engineering.

Nexus Zone operates in **eight service lines**, each backed by at least one shipping product:

Gamer platforms | Digital marketing | Project delivery | Social channels | Cybersecurity | AI solutions | Legacy-code modernization | Enterprise code review

MARKET POSITION

Nexus Zone is the first Arab-founded technology house to combine consumer-facing gaming products (Golden Arena, Waraq Reader) with enterprise-grade engineering tooling (LucidCode, Laundry) under a single Charter. A single customer can build a game, secure a website, modernise legacy code, and receive an Arabic radiology report — from one origin.

Public site: `61465.github.io/game-zone-hub`

Founder: game_nexus_zone.

LucidCode V0.5.0 · PROGRAM VERIFICATION

Version: 0.5.0 **Licence:** Commercial (Nexus Zone 1.0) **Tests:** 330 / 330

Benchmark: F1 = 1.00 on 22 CVE-inspired fixtures

CHAPTER 2 · CODE CONFESSION ENGINE

02 — LucidCode

LucidCode is a developer-experience tool that grants source code cognitive self-awareness. Instead of running static rules over a file, LucidCode makes the code **confess in first person** — and rigorously refutes any confession that cannot be proven.

> *[T1] Suppression @ line 8 — "I confessed: I caught an exception and said nothing. Every scream is muffled the moment it starts. What am I hiding?"*

THE SIGNATURE

THREE-ENGINE ANTI-HALLUCINATION ENSEMBLE

Every confession passes through three independent verifiers before it reaches the user: (1) a deterministic AST re-verifier weighted at 0.90, (2) a sandboxed fuzzer weighted at 0.70, (3) an adversarial LLM devil weighted at 0.40. A Bayesian aggregator produces a calibrated verdict. Hallucinated confessions are logged internally and never displayed.

POSITIONING

- **Semgrep / SonarQube:** static rule match, no reasoning.
- **Copilot:** autocomplete without validation.
- **CriticGPT:** single-model self-critique.
- **LucidCode:** every claim anchored to a real file+line, every fix voted on by three independent engines.

DELIVERABLES

12 CLI sub-commands · VS Code VSIX extension · CodeQL query pack · GitHub App scaffold · arXiv-ready whitepaper · Firecracker sandbox · *Ring* orchestrator (Red-Hat + War-Gaming + Human-Triage roles).

SHIPPING STATUS

> Field-validated on **3 real production repositories**. Benchmark suite of 22 CVE-inspired fixtures achieves **F1 = 1.00**. Commercial licence live under Nexus Zone Commercial 1.0. B2B pricing tier ready.

Landing page: `61465.github.io/lucid/landing`

Laundry V0.3.2 · LEGACY-CODE MODERNIZATION

Version: 0.3.2 **Tests:** 97 / 97 **Codebase:** 62 Python modules

Battle-tested on: Django (2,974 files), Airflow (9,082 files)

CHAPTER 3 · THE LAUNDRY ROOM

03 — Laundry

Laundry is a legacy-code modernization pipeline that **never merges**. It runs twelve specialised agents on a target repository and produces one signed evidence bundle a human engineer can approve — with a full before/after explanation for every unit of code.

> *Legacy code goes in. Evidence comes out. The tool never merges.*

THE SIGNATURE

IMMUTABLE CHARTER · TEN NON-NEGOTIABLE RULES

At session start Laundry loads a Charter of ten runtime-immutable rules — no auto-merge, evidence before edit, judge never writes code, uncited claims dropped, dead code needs two independent proofs, and more. The Charter is stored in a *MappingProxyType* at runtime; any mutation attempt raises *TypeError*. The Charter protects the tool from the tool's operator.

TWELVE SPECIALISED AGENTS

Each agent has one job. The judge never writes code. The updater refuses to run without a passing baseline. The parity agent isolates every patch in a sandbox before accepting it.

Indexer · Security · Business Rules · Dead Code · Tests · Updater · Parity · Architecture · Explainer ⭐ · Passport · Documenter · Judge

VERIFIED FIELD RESULTS

- **Django** (v5.2 · 2,974 Python files) — analysed in **88 seconds**. Surfaced 289 real secrets, 3,607 business rules, 351 dead-code candidates, zero crashes.
- **Apache Airflow** (main branch · 9,082 files) — analysed in **116 seconds**. 486 real secrets surfaced, 5,000 business rules, 2,000 dead-code candidates, six agent-level contradictions correctly escalated.
- **Flask** (85 files, 5.9 s) and **Requests** (37 files, 3.7 s) — control runs confirming false-positive rate on clean code.

COMMERCIAL READINESS

> 21 LLM providers wired (Anthropic, OpenAI, Groq, Cerebras, Gemini, Bedrock, Vertex, and more). 15 integration categories (Slack, Sentry, Jira, Linear, etc.) via 138 documented environment keys. SARIF export lands directly in GitHub / GitLab / Azure DevOps code-scanning surfaces. HMAC signed-override protocol enforces human authorisation before any gate is bypassed.

Interactive live demo: `61465.github.io/lucid/landing/laundry.html` (runs against *psf/requests v2.7.0*, May 2015).

NEXUS-AI V0.7 · 51-AGENT ORCHESTRATION FABRIC

Agents: 51 **Models:** 8 LLMs behind one facade **Skills:** 1,580 curated **Protocol:** MCP JSON-RPC 2.0

CHAPTER 4 · THE 134-AGENT FABRIC + CYBER GROUNDING GATE

04 — NEXUS-AI

Most teams ship one or three agents. NEXUS-AI ships a **134-agent catalog** — thirteen in active production rotation, the remainder in a specialist reserve pool. The value is not in the raw count; it is in the coordination layer (Meta-Router + Verified Runner + hash-chained audit) and the **Cyber Grounding Gate** — a five-verifier LLM-hallucination filter that has stripped **724 fabricated claims** across **770 gate calls** in production, backed by **173 curated knowledge rules** and a **166-CVE offline cache**. Doctor status: **6/6 GREEN · 179 tests passing**.

THE SIGNATURE

> **META-ROUTER WITH 100% TASK CLASSIFICATION ACCURACY**
>
> Every request enters through the *Meta-Router*, a rule-based classifier that maps the request to a task type (code_gen · reasoning · web_search · vision · osint · writing …) and then dispatches to the correct agent and the cheapest capable model. Verified 100% classification accuracy on a 13-request live test suite spanning real production traffic.

VERIFIED RUNNER — THE TRUTH LAYER

Every agent response is filtered through three tiers before it leaves the fabric: (1) Tier-0 schema check, (2) Tier-1 NLI self-check, (3) 13-signature injection detector. Only fully-cleared responses reach the caller.

COMPANY ORCHESTRATOR

For composite tasks, the Company Orchestrator decomposes the request into sub-tasks, dispatches them in parallel across specialised agents, verifies each result, and merges. Functionally, one call yields the output of an engineering department — at LLM latency.

> **PRODUCTION FOOTPRINT**
>
> NEXUS-AI is the shared brain across **midcine** (radiology inference orchestration), **thawani.cc** (customer-response automations), and internal engineering workflows. The Meta-Router prefers free-tier providers (Naraya · Groq · Cerebras · Gemini · OpenRouter — sixteen providers total), which drives real per-request cost close to zero. Every cyber-agent output is post-processed through the grounding gate: browser-security invariants, OAuth

> flow rules, CVE cross-check, identifier grep-existence, and chain-step reproducibility labelling — hallucinated claims are stripped before they reach the user.

midcine ARABIC RADIOLOGY AI · RIS/PACS

Target: Hospitals in MENA **Read time:** 24-slice CT in ~69 s **Confidence:** 92%

Output: DICOM SR-signed Arabic report

CHAPTER 5 · RADIOLOGY BEHIND THE FIREWALL

05 — midcine

Most radiology-AI products run in US-hosted clouds. MENA hospitals cannot legally send patient scans there. midcine is designed to run **inside the hospital firewall** — no external network access unless explicitly authorised — and to write reports directly in Arabic, signed as standard DICOM SR objects.

MEASURED PERFORMANCE

- Reads a 24-slice CT into a structured report in **~69 seconds**.
- AI confidence: **92%**.
- Guardian review reproduced an ACR-grade subarachnoid haemorrhage diagnosis.
- Runs on 8 GB RAM minimum, 16 GB recommended.
- Ten micro-services in a single pnpm/turbo monorepo.

THE SIGNATURE

> **VISION + CLINICAL LLM ENSEMBLE**
>
> midcine does not rely on a single model. Its pipeline chains a *Vision Classifier* (lesion class), a *Detection Model* (bounding boxes), a *Segmentation Model* (volume maps), a *Vision-Language* reader (radiological description), and a *Clinical LLM* (final report composition). Each model cross-verifies the previous one; disagreements route to a human reviewer.

> **OFFLINE-FIRST ROADMAP**
>
> Egyptian hospital networks are unreliable. midcine ships with Ollama fallbacks so an entire study can be read without any Internet connection. Optional premium services (TotalSegmentator, external PACS bridging) are opt-in.

Live demo: `ame.tail19ddab.ts.net:8445/reports` (private Tailscale Funnel · synthetic patients).

Source showcase: github.com/61465/midcine

MEDNEXA HEALTHCARE INTEGRATION PLATFORM · HIS × LIS × PACS

Target: Egyptian mid-market hospitals **Compliance:** PDPC Law 151/2020 + HIPAA-aligned

Status: Session 01 complete · pilot-ready **Engine tests:** 128 / 128 passing

Source showcase: github.com/61465/MEDNEXA **Pilot & DPA inquiries:** `abdarahman10555@gmail.com`

CHAPTER 5.5 · CONNECTING HEALTHCARE

05.5 — MEDNEXA

Where **midcine** owns the reading side of radiology, **MEDNEXA** owns the **wiring** between the hospital's existing HIS, LIS, PACS, and clinical workflows. Egyptian mid-market hospitals typically run three or four disconnected vendor systems; MEDNEXA stitches them together with real HL7v2 / FHIR R4 / DICOM pipes, PDPC-compliant consent management, and a hash-chained audit log that makes any after-the-fact mutation detectable.

WHAT THE ENGINE ACTUALLY SHIPS

- **FastAPI engine** (Python 3.12) with 80+ REST routes.
- **Real DIMSE SCP** receiving `C-STORE` and `C-FIND Modality Worklist` on port `11112`.
- **HL7v2 parser** covering `ADT` / `ORM` / `ORU` / `OUL` / `MDM`, piping into FHIR R4 builders (`Observation`, `DiagnosticReport`, `ServiceRequest`, `Patient`, `Practitioner`).
- **DICOM in-browser viewer** on Cornerstone.js for sample CT studies.
- **Report state machine**: `draft → signed → finalized → amended` with full audit trace at every transition.
- **Hash-chained audit log** (SHA-256 `prev_hash + hash`) — any mutation after the fact is detectable.
- **MFA TOTP** (RFC 6238) with AES-256-GCM-encrypted secrets at rest.
- **Refresh-token rotation** (SHA-256-hashed), sliding-window per-IP + per-user rate limits.
- **Bilingual UI**: Arabic RTL + English LTR, Clinical Editorial design system.
- **nginx 1.27 reverse proxy** with CSP / HSTS / X-Frame-Options.

THE SIGNATURE — HASH-CHAINED AUDIT

Every state transition in the engine (login, report sign, DICOM store, consent grant, export) writes an entry carrying both its own SHA-256 and the previous entry's SHA-256. Rolling back the audit log requires rolling back every subsequent entry too — which is detectable at verification time. This is the core of the PDPC Law 151/2020 defensive posture.

FOUR-ROLE DEMO MATRIX

> Pilot deployments expose Admin · Radiologist · Doctor (referring) · Patient. Scoped-ACL enforcement means the referring doctor sees only the patients a patient has consented to for them, with admin approval also required — a two-key gate per access, logged.

DELIBERATELY NOT SHIPPED PUBLICLY

The strategy, 15-slide pitch deck, 36-month × 3-scenario financial model, DPA (PDPC Law 151/2020-aligned) + SLA + SOW templates, 30-question discovery questionnaire, ROI calculator, and the engine source itself live under commercial license and ship under NDA for pilot evaluations. The showcase repository on GitHub contains the README, brand, and compliance posture only.

thawani.cc WHATSAPP COMMERCE SAAS

Market: Saudi + Gulf **Status:** Live paid tenants **CR:** 3453105300 **VAT:** 311174977200003

Live: thawani.cc **Backup landing (if domain is down):** 61465.github.io/thawanidemo

Source: github.com/61465/Thawani

CHAPTER 6 · WHATSAPP AS A STOREFRONT

06 — thawani.cc

Every WhatsApp-commerce project hits the same wall: **Meta bans sessions**. Meta's own baseline ban rate is 0.5%. At scale, that means one tenant lost every day.

thawani.cc treats the ban problem as an engineering problem, not an operational one. The response is a **26-layer defence stack** operating in parallel.

THE SIGNATURE

> **26-LAYER DEFENCE STACK**
>
> Stealth Mode v2 · Behavior Analyzer + Gatekeeper (a seven-rule 0–100 scoring model) · Phase-2A humanisation · adaptive rate limiting · reputation tracking · phone-uniqueness enforcement · and twenty other independent layers. Compound effect: the observed ban rate drops from Meta's 0.5% baseline to **0.1%** — a five-fold improvement across live tenants.

SIX VERTICAL MODULES · ONE SHARED KERNEL

Bookings · Dine-in QR · Accommodation · Gaming Topup · Restaurant Reservations · Lamah Calendar

SALLA INTEGRATION

Salla is Saudi Arabia's largest e-commerce platform (55,000+ MENA stores). Phase 1 is live: OAuth 2.0 + Webhooks + product sync + disconnect. Every Salla store can convert to a thawani tenant with a single button.

> **COMMERCIAL MODEL**
>
> SAR 50 / store / month via Meta WhatsApp Cloud API in Tech-Provider mode. Even a 1% Salla penetration yields SAR 27,500 monthly recurring revenue. Six-vertical shared-kernel design keeps unit economics healthy across scale.

ARIA v9.0 / GZP-LLM

SOVEREIGN AI COMPANION

Base: Qwen2.5-3B (QLoRA r=64) + Qwen2.5-1.5B (chunked variant)

Tuning: QLoRA + SFT (3 epochs) + DPO (2 iterations)

Data: 5,488 bilingual samples (SFT 2,918 · DPO 487 · eval 154 · finetune 1,929)

Compute: Kaggle T4/P100 GPU (QLoRA pipeline) **Design goal:** one human

CHAPTER 7 · A MODEL FOR ONE HUMAN

07 — ARIA v9.0

Most AI models are "for everyone." ARIA is deliberately **not**. ARIA is designed to grow through an exclusive relationship with a single human — it has memory, a personality, autonomous background cognition, and a curiosity engine that asks real questions to fill its sensory blind spots.

> *Not a chatbot. An intelligence that knows its owner.*

THE SIGNATURE

CURIOSITY ENGINE

Every hour, a background "curiosity" pass reviews the day's conversations, identifies gaps in the model's world model, and formulates targeted questions the human is prompted to answer. Over time ARIA **knows its owner better** than any general-purpose assistant.

STRATEGIC VALUE

ARIA is not currently for sale. It is a **life pattern**. Once the internal loop is fully stabilised, the underlying infrastructure will be released as a hosted SaaS so any individual can build their own sovereign companion.

CURRENT STATE

- ARIA v9.0 base runtime live: FastAPI :9090 · persistent aria_brain.db (~232 MB) · 8-mind cascade over 8 free LLM providers.
- GZP-LLM training stack: 5,488 curated bilingual instruction samples ready; Kaggle QLoRA notebook prepared (Qwen2.5-3B, r=64, SFT→DPO).
- **Coder variant shipped:** `coder15b_final` — merged LoRA adapter (17.4 MB safetensors + tokenizer + inference.py + chat template).
- Autonomous stack: curiosity engine · closed-room reflection (6 fractal sub-rooms) · weekly self-improvement scheduler · real-time gaming overlay.

- Long-term memory persisted locally. Arabic + English bilingual.

NEXUS Vigil™ (Smart Fleet AI)

ABANDONED-VEHICLE INTELLIGENCE

Market: US municipalities **Coverage:** 6 states + 40 cities **Endpoints:** 44 REST
Database: 97 MB live SQLite (WAL)

CHAPTER 8 · EVERY ABANDONED CAR HAS A STORY

08 — NEXUS Vigil™

US municipalities spend millions removing abandoned vehicles. They do not know the vehicle's age, its owner history, or whether it is subject to manufacturer recall. NEXUS Vigil™ resolves all three in a single API call.

THE SIGNATURE

MAPILLARY GRAPH API V4 + NHTSA VPIC / RECALLS

The product fuses two of the strongest open datasets in North America: *Mapillary* (crowdsourced street imagery at scale) and *NHTSA* (vehicle-manufacturer and recall data). Vigil captures a plate, resolves make/year, checks recall status, and opens a case file — in seconds.

PROPRIETARY PRODUCT

Full intellectual-property ownership. Municipalities buy subscription access; live SQLite database auto-updates. Coverage: Massachusetts, Connecticut, Rhode Island, New Hampshire, Vermont, Maine — plus 40 individual cities. Roadmap: EU municipalities.

CHAPTER 9 · THE SECURITY STACK

09 — Applied-Security Products

Four products operate together as a single security fabric:

NEXUS SecOps + NEXUS_AUDIT Arsenal

RED / BLUE / AI / OSINT

32 registered security tools (Amass, Nuclei, Metasploit, CodeQL, garak, PyRIT, Shodan, Censys, and more) plus 4 live threat-intelligence feeds (GreyNoise · Leakix · urlscan · PublicWWW) exposed behind seven REST endpoints. A single command `nexus-audit <target>` wraps 10+ SAST/DAST tools with automatic language detection and language-specific rule packs.

NEXUS Firm

POLICY-GOVERNED ENGINEERING CONTROL PLANE

One architectural rule: *LLM proposes → tools produce evidence → policy decides → agent executes within capability bounds.* Inter-agent handoffs are typed artifacts, never prose. Every state mutation is a validated, logged tool call.

OPA / Rego · MCP Gateway · Firecracker · gVisor · Capability Lattice · Evidence-WORM Memory

SENTINEL Enterprise + sentinel-cli

WINDOWS SECURITY SUITE

Windows SIEM + SOAR + ML in one bundle. For organisations that need endpoint protection on Windows without paying CrowdStrike enterprise pricing. Terminal companion (`sentinel-cli`) ships as a stand-alone open showcase repository.

VulnAdvisor Pro

LOCAL VULNERABILITY TRIAGE & ADVISORY ENGINE

A local engine that ingests findings from Semgrep, Nuclei, and CodeQL and re-ranks them by **real exploitability**, not theoretical severity. Saves security teams days of noise triage per week.

CHAPTER 10 · GAMING UNIVERSE

10 — Consumer Gaming Platforms

Six shipped consumer products serving the Arabic gaming audience:

Golden Arena 518

AI-POWERED FANTASY BATTLE PLATFORM

A fantasy arena that pits characters from anime, video games, and film. AI analyses each character's abilities; an ELO system ranks them; an interactive tier list surfaces the meta.

URL: `61465.github.io/game-zone-golden`

Game Zone Archive

ENCYCLOPAEDIA + AI DNA ANALYZER

200 characters × 200 games × 60+ franchises. An AI "DNA" analyser produces personalised game recommendations.

URL: `61465.github.io/game-zone-archive`

The Last Save

INTERACTIVE NARRATIVE GAME

An interactive narrative experience about memory and identity. Every decision alters the ending.

URL: `61465.github.io/game-zone-game`

GZ Chat Pro

ARABIC-FIRST GAMER COMMUNITY PLATFORM

Voice rooms, embedded mini-games, rank & XP system — an Arabic-first alternative to Discord for the MENA gamer community.

URL: `61465.github.io/GAME-ZONE-CHAT`

Waraq Reader

EDUCATIONAL TRANSLATION PLATFORM

Multi-language reading with four study modes and Arabic AI translation support. Positioned as an educational reader for MENA students.

URL: `61465.github.io/read`

Golden Arena Configurator

DECISION-ANALYSIS TOOL

Compares games and hardware using radar charts and structured text analysis.

URL: `61465.github.io/game-zone-CONFIGURATOR`

CHAPTER 11 · SPECIALISED VERTICALS

11 — Specialised Products

Zwaya Fit v0.3.0

WHATSAPP AI HEALTH COACH

A full AI health coach delivered inside WhatsApp. Baileys + Groq stack. 16 REST endpoints. Groq-Vision meal analysis. Auto-generated weekly weight-tracking PNGs. Eleven achievement badges. Six context-aware alert types on a 10 AM / 6 PM scheduler. Exercise library with TTS audio. Family mode. Forty-feature admin dashboard.

GZ CyberRange v0.1

FINTECH TRAINING LAB · 30 PLANTED VULNERABILITIES

A cyber range with thirty planted vulnerabilities spanning OWASP Top-10 · fintech-specific · Next.js SSR · API misc. Coverage validated at **86%** by an automated NEXUS attack loop. Positioned for corporate security-team training.

Restaurant Management Suite

NEXT.JS + PRISMA + VERCEL

End-to-end restaurant operations. Menu with 200+ items. Deployed for one MENA operator. Prisma schema ready for multi-tenant rollout.

V24 Operations PWA

TALABAT + INSTASHOP DASHBOARD

Multi-channel operations dashboard for MENA food-delivery brands. Shift KPIs. Voice dictation. Supabase sync.

CHAPTER 12 · PUBLISHED RESEARCH

12 — Research Portfolio

SSRN 6870178 · MIT Licence

Ariadne — Adaptive HITL Defence for AI Agents

A four-pillar defence framework against indirect prompt injection targeting autonomous AI agents: (1) least-privilege scoping, (2) tamper-evident observability, (3) synthetic deception, (4) human-gated remediation.

> **REPRODUCIBLE RESULT**
>
> 8,000-session AgentDojo benchmark. **90.6% adversarial containment** at 5.9% false-positive rate. Least-privilege pillar alone reaches 56.5% containment at **0% FPR**.

SSRN 6602478 · IEEE TIFS extended bundle in preparation

Oblivion Gate — Zero-Trace Fragmentation in Distributed Data Architectures

Three novel cryptographic protocols (BDSP · STCHP · HS-CPP) for zero-trace fragmentation of sensitive data plus decoy-based threat saturation. The core innovation: instead of encrypting the data, shatter it into individually meaningless fragments, distribute them, and populate the space with plausible decoys that saturate any adversary's search budget.

arXiv-ready · Nexus Zone Commercial 1.0

LucidCode — Program Verification with Grounded Anthropomorphic Interfaces

The academic paper behind LucidCode. Unifies compositional reasoning, adversarial mutation testing, provenance tracking, citation enforcement, and a zero-LLM truth path. Central philosophical claim: *you do not need a language model to know the truth — you only need one to describe it.*

CHAPTER 13 · COMPETITIVE RESULTS

13 — Kaggle Track Record

10,187 teams competed across four live Kaggle tracks. Independent submissions against university and corporate teams. Verified final scores (JED competition finalized 2026-09-03):

88.35	0.936	1.78	30.14
JED — Top 21.3% (#892 / 4,187)	RSNA AUC — Top 16.7% (#336 / 2,013)	ARC-AGI-3 — Top 22.6% (#549 / 2,424)	ARC-AGI-2 — Top 32.2% (#503 / 1,563)

THE DISTILLED PLAYBOOK

> **PLAYBOOK_KAGGLE.MD — VERBATIM-FIRST**
>
> Extracted from twelve failed modification attempts across three competitions: **clone the strongest public notebook verbatim first, tune one knob per submission, never combine changes**. Eleven of twelve exploratory modifications regressed. The playbook is now a shared internal artefact for future Kaggle work.

PUBLISHED WRITEUPS

- **ARC-AGI-2 Paper Track:** "The Symbolic Ceiling on ARC-AGI-2" — a novel application of Friston's Free Energy Principle to ARC; ρ(τ) computed on 320 tasks.
- **Pokémon TCG Strategy:** An α-Rank statistical study with Wilson confidence intervals on 200 games per cell, plus a five-signature strategic framework.

CHAPTER 14 · RESPONSIBLE DISCLOSURE

14 — Applied-Security Track Record

Twenty+ engagements across Google ASR / OSS VRP / Cloud VRP, HackerOne, Bugcrowd, Trend Micro ZDI, GitHub Security Advisories, and internal ranges. October 2026 added the APDS methodology submission to Google Android ASR (CVSS 10.0 Critical, in panel review), two accepted GHSAs (Trigger.dev CVSS 9.6 + Documenso 7.6 in triage), and five further Google VRP tickets. Earlier accepted: Google Gemini CLI S0/P0, Syfe CVSS 8.6, Gearset P3.

PROGRAM	PLATFORM	SEVERITY	VULNERABILITY	STATUS
Google Android ASR	Issue 571135577	CVSS 10.0	**APDS methodology + CVE-2024-40658 symmetric sibling.** First formal application of Asymmetric Patch Detection for Symmetric APIs. Un-patched sibling of CVE-2024-40658 byte-identical across 6 AOSP branches (android14-qpr3 → android17-release) for 25 months of ASB silence. Live binary verification from a Google-published Android 16 QPR1 emulator image (security patch 2026-01-05). 3/3 reproduction runs captured. Open-sourced the reference tool as `apds-hunt` within 24 h of the submission commitment.	**In Panel Review**
Trigger.dev	GHSA-65rr-73jq-6qhf	CVSS 9.6	Accepted by @ChrisArderne on 2026-10-07. 90-day disclosure clock running.	**Accepted**
Documenso	GHSA-2rp7-mrh6-v9m2	CVSS 7.6	Rejected-recipient field forgery across 3 tRPC routes. First-ever GHSA on the Documenso repo.	**In Triage**
Google Cloud VRP	Issue 569156158	CVSS 9.9	Vertex AI pickle cross-tenant RCE — `pickle_object_gcs_uri` free-form path → Vertex Agent Engine arbitrary code execution across tenant boundaries.	**AI-triage done**
Google OSS VRP	Issue 569269080	CVSS 9.6	Google ADK — AgentRegistry sends Google Cloud ADC OAuth token to ANY registered MCP URL. P1 upgraded by panel in 5 minutes.	**P1 Upgraded**
Google Cloud VRP	Issue 569402823	CVSS 9.9	Agent Builder — RLHF poisoning via `feedbackEntries` + memory-banks BOLA.	**AI-triage done**
Google Cloud VRP	Issue 569575994	CVSS 10.0	Agent Builder — `CodeExecutionMetric.custom_function` arbitrary Python RCE via 5 endpoints; 6 code-	**In Triage**

PROGRAM	PLATFORM	SEVERITY	VULNERABILITY	STATUS
			execution primitives; GDPR/HIPAA bypass via 24-region sandbox-shopping.	
Google OSS VRP	Issue 563082721	S0 / P0	Gemini CLI chained RCE — 6-defect zero-click supply-chain chain, CVSS 10.0. Security engineer took over, linked to a blocked-by bug. Panel decision in 2–3 weeks; expected reward band $30K–$75K.	**Accepted**
Syfe	HackerOne #3914035	CVSS 8.6	Web Cache Deception (CWE-524)	**Accepted**
Gearset	Bugcrowd 0e7d2c90	P3	Excessive GitHub-OAuth scope (repo + write:repo_hook) on staging.claytonapp.com	**Accepted**
Cal.com	Bugcrowd 30c5dfce	CVSS 9.1	Unauthenticated zero-click PII disclosure in `getBookingForReschedule` — two compound defects (authorization gated behind seat check + inverted attendees ternary). Two public `getServerSideProps` sinks. **Live-verified on `app.cal.qa`**: a fresh browser context with zero cookies received an HTTP 200 whose HTML body contained the full `attendees[]` record (email + name + `bookingSeat`) plus `responses`, `customInputs`, `location`, and times. Delivered with a video PoC, a proof panel, the verbatim response body, and a tested private patch.	**Submitted**
OpenAI Safety	Bugcrowd cf0acebe	P1	Cross-tenant PII leakage across 7 production models (5 chained failure modes — HIPAA, GLBA, attorney-client, SEC-material, HR insider). Silent-execution deception proven.	**Submitted**
Trend Micro ZDI	NexusZone0001	CVSS 9.0	Windows fcon.dll + wosc.dll signature-verify gap via WNF cache poison — proof the NEXUS pipeline works end-to-end on Windows binaries.	**Submitted**
xAI Grok	HackerOne #4024682	—	Cross-model system-prompt disclosure on Grok API. Canary 5/5 ASR on grok-4.20-non-reasoning + grok-build-0.1. 7 data classes leaked (PHI / GLBA / SEC / AWS / legal / M&A / HR). Tool-chain exfil proven; distinct from Adversa Aug '26.	**Vendor Review**

PROGRAM	PLATFORM	SEVERITY	VULNERABILITY	STATUS
Shopify	HackerOne #3919475	CVSS 8.7	Unified 7-vector chain (UCP SSRF + GCP trace leak + OTLP unauth + tenant spoof + schema oracle)	**Triage**
AXIS OS	Bugcrowd 8a7425f0	P2	ACAP D-Bus wildcard bypass of CVE-2025-5452 (firmware-validated on M1075-L 12.11.77)	**Submitted**
eToro	Bugcrowd f83be6be	P1	1-click ATO on Delta Desktop (6 chained bugs, live PoC)	**Submitted**
Okta AtSpoke	Bugcrowd 6d26be3a	P1	Application-wide CSRF via hardcoded `XSRF-TOKEN=JustAskSpoke!`	**Submitted**
Magic Labs	Bugcrowd 696307c8	P3	reCAPTCHA bypass + user enumeration (13 staff accounts confirmed, including CEO and CTO)	**Submitted**
GZ CyberRange (self-built lab)	Internal	—	30-vulnerability fintech training lab; 86% hunter coverage validated	**Live**

Hire-signal open-source contributions

Two contribution-first PRs opened at engineering companies with public "we hire from our contributor pool" culture. Each PR was designed to sit at the intersection of a real user-facing bug and a technical difficulty bar the maintainers care about, and each was accompanied by a 4-element response protocol (acknowledge / state change / evidence / offer flexibility) developed and locked over the last two sessions.

PostHog · posthog-js · PR #5083 · Sep 2026

Opt-in bot-detection heuristics (issue #2921)

Regex-based detector of impossible Chrome patch/build tuples embedded in user-agent strings — closes a long-standing PostHog issue about a Chrome-forged bot family that other detectors miss. Delivered with a **5.1×–10.9× perf leap** over the naive baseline, **32 new unit tests** pushing coverage on the new file to **98.32%**, the full **2,095 / 2,095** core suite green, and a **5/5** Wiz-scan pass. Blocking review from @marandaneto surfaced a live-Chrome edge case (Chrome 154.0.8037.58 and 155.0.8059.12 already shipping); fixed surgically by removing the build-ceiling rule the same session and posting a 4-part reply within the 4-hour SLA.

Supabase · storage · PR #1439 · Sep 2026

Full UTF-8 object keys + invisible-glyph reject list

Revives the direction of PR #875 with a scope-controlled patch: expands the allowed object-key character set to full UTF-8, rejects **17 zero-width / bidi / C1-control codepoints** to prevent homograph attacks, and fixes two latent bugs discovered in the process (backslash escape, path traversal). **76 / 76** tests pass, **10,000** Hypothesis-driven fuzz cases pass, plus a benchmark run. When Supabase's `depthfirst-app` bot flagged four additional Unicode codepoints, all four were added to the reject list with tests in the same session. Staff maintainer **@ferhatelmas** engaged twice; awaiting a decision on how to sequence the follow-up work.

CHAPTER 15 · ROADMAP AHEAD

15 — Roadmap

NEXT 90 DAYS

- Onboard the first three paying pilots for **Laundry** at USD 499 / month per repository.
- Publish first hospital pilot for **midcine** in Egypt.
- Complete peer-review preparation for the **LucidCode** paper.
- Announce the Arabic-first LLM (Nexus Zone AI model — evaluation phase).

NEXT 6 MONTHS

- Hosted multi-tenant dashboard for **Laundry**.
- SOC 2 certification path opened to unlock enterprise sales.
- Ten paying customers · USD 30K ARR target.
- Publish the **Ariadne** IEEE TIFS extended bundle.

NEXT 12 MONTHS

- Nexus Zone at USD 100K ARR.
- First pre-seed or seed round.
- Team expansion to three engineers.
- ARIA v10 with open-source companion infrastructure.

> *A product is not judged by the demo. It is judged by the reproducible run.*

Engage With Nexus Zone

Every product in this catalog is real, published, and independently verifiable. Requests for evaluation copies, pilot programmes, licensing conversations, or research collaboration are welcomed.

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Response within one business day.

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